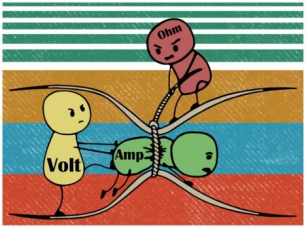
Wk15L1: Wrap-up

Giles Blaney, PhD



Tufts UNIVERSITY | School of Engineering

Mon. Dec. 8

Course Objective

By the end of this course students will be able to **analyze**, **design**, and **build** basic electrical circuits commonly encountered in biomedical devices. These circuits may include components such as resistors, capacitors, and op-amps. From **analysis**, students can identify voltages and currents within a circuit and understand the different behaviors of circuits in terms of steady-state, transient, and sinusoidal oscillations. For **design**, students will be comfortable with implementing and understand voltage dividers, Wheatstone bridges, and basic op-amp configurations. Finally, when **building** circuits students can identify the aforementioned components and implement a circuit diagram using these real world components on a breadboard.

- Analyze, design, and build basic electrical circuits including:
  - **Resistors** Wk2-Wk14
  - **Capacitors** Wk8-Wk14
  - **Op-amps** Wk4-Wk14
- Analyze voltages and currents in:
  - **Steady-state** Wk2-Wk7
  - **Transient** Wk8-Wk9
  - **Sinusoidal oscillations** Wk10-Wk14
- Design and understand:
  - **Voltage dividers** Wk3-Wk14
  - **Wheatstone bridges** Wk3-Wk5
  - **Basic op-amps** Wk4-Wk14
- Build a circuit diagram on a **breadboard** Demos

Quizzes and the Final

Quiz	Chapter(s)	Topic(s)
I	1	Terminology & Ohm's law
II	2-3	Equivalent circuits & Voltage dividers
III	4	Circuits with op-amps
IV	5-7	Circuits with capacitors and/or inductors
V	9 & 13	Transfer functions

- Quizzes II, III, IV, and V will have questions associated with them on the final.
- Quiz I will have an **optional** final question associated with it.
- An additional **optional** final extra credit question will be associated with the Fourier Transform topic—this extra credit will apply to the midterm.

In all the final will be 4 required questions and 2 optional questions.

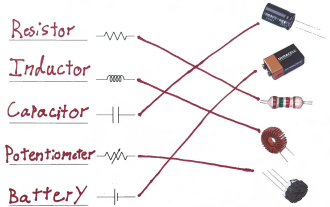
Quiz I

BME 0010 — Electrical Circuits & Biomedical Applications — Fall 2025 Quiz I

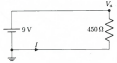
Name: Solution

Question 1: (10 pts) Match each of the circuit diagram components to their corresponding photo.

Question 2: (10 pts) Write the name of each circuit diagram component on the line next to its symbol.



Question 3: (20 pts) Given the circuit diagram below, find I and V_a.



$\boxed{V_a = 9V}$

$I = \frac{-V_a}{450\Omega} \Rightarrow \boxed{I = -0.02\,A}$

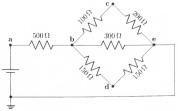
Question 4: (10 pts) Given the above circuit diagram, find the power delivered by the battery.

$P = IV = (0.02)(9) \Rightarrow \boxed{P = 0.18\,W}$

Quiz II

BME 0010 - Electrical Circuits & Biomedical Applications — Fall 2025 Quiz II

Name: Solution



Consider the circuit above for the following questions.

Question 1: (10 pts) Find the equivalent resistance between nodes **b** and **e** (i.e., R_{be}).

$R_{be} = (100+200) \| (300) \| (150+150)$

$= 300 \| 300 \| 300 \Rightarrow \boxed{R_{be} = 100\,\Omega}$

*Note: Accept $R_{ae} = 600\,\Omega$

Question 2: (20 pts) Find the voltage between node **a** and node **b** (i.e., V_{ab}).

$V_{ab} = \frac{500}{500+100}(24) \Rightarrow \boxed{V_{ab} = 20\text{ V}}$

Question 3: (20 pts) Find the voltage between node **c** and ground (i.e., V_c).

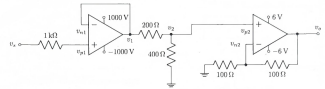
$V_{ae} = 24 = V_{ab} + V_{be} \Rightarrow V_{be} = 4\text{ V}$

$V_c = V_{ce} = \frac{200}{100+200}(4) \Rightarrow V_{ce} = \boxed{V_c = 2.\overline{6}\text{ V}}$

Quiz III

BME 0010 – Electrical Circuits & Biomedical Applications Fall 2025 Quiz III

Name: Solution



Consider the circuit above for the following questions.

Question 1: (10 pts) Find v_1 in terms of v_s.

$i_{p1} = 0 \Rightarrow V_{1k\Omega} = 0 \Rightarrow \boxed{V_1 = V_s}$

Question 2: (15 pts) Write an equation for v_2 in terms of v_s.

$V_2 = \frac{400}{200+400} V_1 = \frac{2}{3} V_1 \Rightarrow \boxed{V_2 = \frac{2}{3} V_s}$

Question 3: (20 pts) Write an equation for v_o in terms of v_s.

$V_{n2} = \frac{100}{100+100} V_o \Rightarrow V_{n2} = \frac{1}{2} V_o$

$V_{n2} = V_{p2} = V_2 = \frac{2}{3} V_s \Rightarrow \frac{2}{3} V_s = \frac{1}{2} V_o$

$\Rightarrow \boxed{V_o = \frac{4}{3} V_s}$

Question 4: (5 pts) Given the power rails, for which range of values of v_s is your equation for v_o valid?

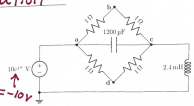
$-6 \le V_o \le 6 \Rightarrow -6 \le \frac{4}{3} V_s \le 6$

$\Rightarrow \boxed{-4.5\,V \le V_s \le 4.5\,V}$

Quiz IV

BME 0010 – Electrical Circuits & Biomedical Applications — Fall 2025 Quiz IV

Name: Solution



Consider the circuit above for the following questions and use $\omega = 420\ \text{rad s}^{-1}$.

Question 1: (5 pt) Find the linear frequency (f).

$f = \frac{\omega}{2\pi} \Rightarrow \boxed{f = 67\ \text{Hz}}$

Question 2: (5 pts) Find the impedance of the inductor (Z_L).
Report your answer in rectangular form.

$Z_L = j\omega L = j(420)(0.0024) = \boxed{j\,\Omega}$

Question 3: (5 pts) Find the impedance of the capacitor (Z_C).
Report your answer in rectangular form.

$Z_C = \frac{1}{j\omega C} = \frac{1}{j(420)(1200\times10^{-6})} = \boxed{-2j\,\Omega}$

Question 4: (15 pts) Find the impedance between nodes a and c (Z_{ac}).
Report your answer in rectangular form.

$Z_{ac} = \left(\frac{1}{10} + \frac{1}{-2j} + \frac{1}{10}\right)^{-1} = \left(1 + j/2\right)^{-1} = \frac{1}{1+j/2}\left(\frac{1-j/2}{1-j/2}\right)$

$= \frac{1-j/2}{5/4} \Rightarrow \boxed{Z_{ac} = \left(\frac{4}{5} - j\frac{2}{5}\right)\Omega}$

Question 5: (20 pts) Find the voltage phasor at node c (V_c).
Report your answer in polar form.

$\vec{V}_c = \frac{Z_L}{Z_{ac}+Z_L}\vec{V}_b = \frac{j}{4/5 - j2/5}(-10) = \frac{-50j}{4+3j}\left(\frac{4-3j}{4-3j}\right) = \frac{-200j-150}{25}$

$= \underline{[-6-8j]\ \text{V}}$

$\phi = \tan^{-1}(8/6) = 0.93$ rad

$\Rightarrow \phi = 0.93 + \pi = 4$ rad

$\sqrt{6^2+8^2} = \underline{10\ \text{V}}$

$\Rightarrow \boxed{\vec{V}_c = 10e^{j4}\ \text{V}}$

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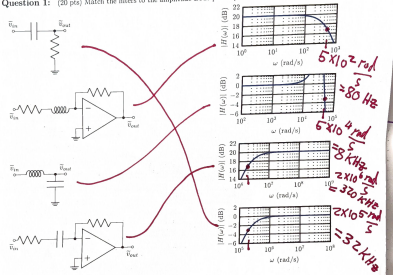
Quiz V

BME 0010 – Electrical Circuits & Biomedical Applications Fall 2025 Quiz V [Last Quiz \o/]

Name: Solution

Question 1: (20 pts) Match the filters to the amplitude Bode plots by drawing a line connecting them.



Question 2: (10 pts) Choose one of the filters above and estimate its cut-off (*i.e.*, corner) frequency in Hz (f_c) based on the Bode plot.

Question 3: (20 pts) Choose one of the filters above and derive its transfer function ($H(\omega)$). For every component assume: $R = 5\,\Omega$, $C = 1\,\mu$F, & $L = 1$ mH. *Show all of your steps.*

See Lecture Notes

Reminders

- Final: Friday, December 12
  3:30 PM - 5:30 PM
  JCC 180 (usual classroom)